Viplove Raj Sharma

Associate Director at Great Learning

- 14+ years of experience in designing, developing and delivering analytics and data science solutions for businesses
- Consulting senior management and leadership across geographies, industries, and functions
- Earlier at the Royal Melbourne Institute of Technology (RMIT) Melbourne, and Mu Sigma, Bangalore
- Leading product and delivery of Great Learning’s data programs – analytics, data science and AI

From Insights to Impact

Transforming Industries with Data Science

Learning Objectives

- Develop an understanding of what is an “insight” for a business
- Get exposure to how problems exist in the real world
- Get a view on how data science can solve those problems
- Get initiated on what it takes to navigate that journey from insights to impact

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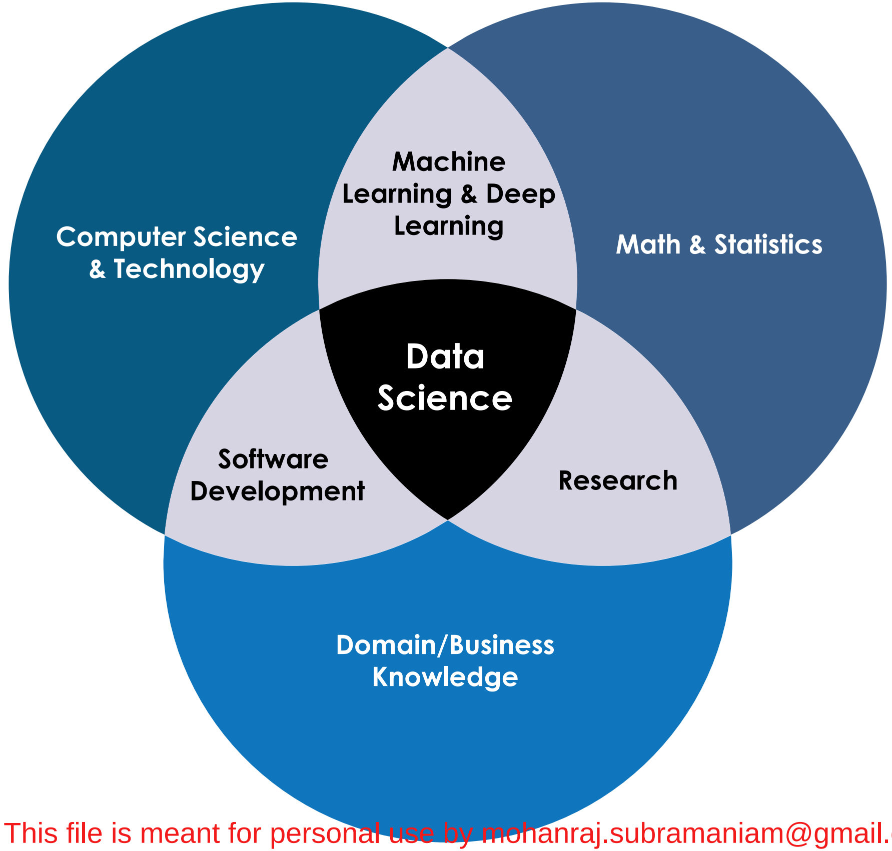
Revisiting definition of Data Science



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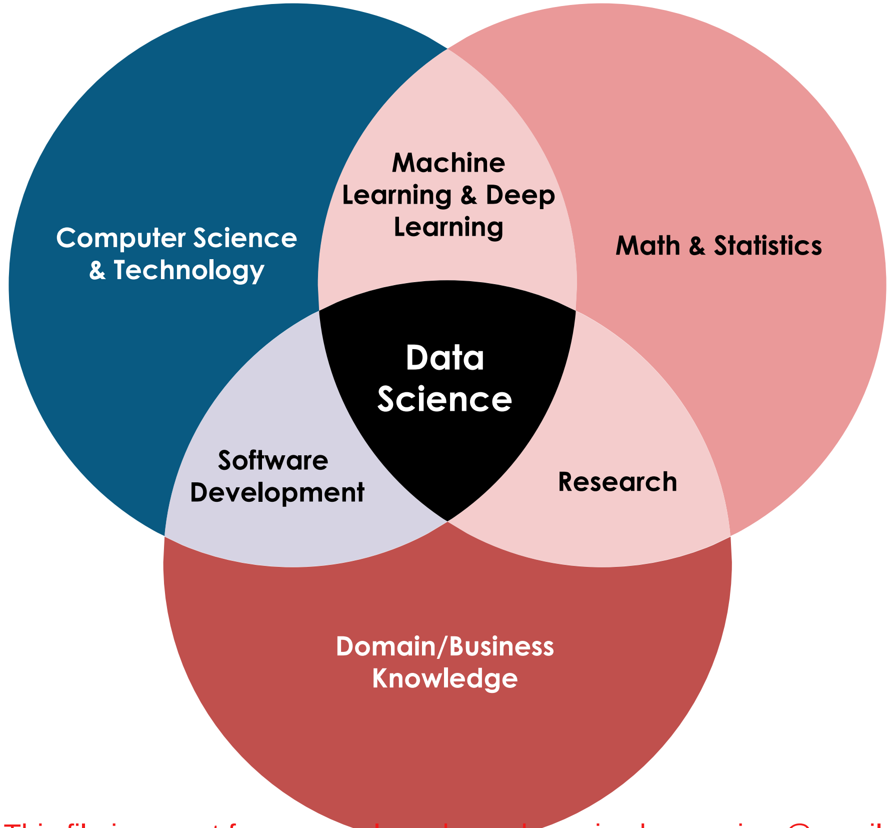
Revisiting definition of Data Science – today's focus



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Problem Space - Retail Industry

retail

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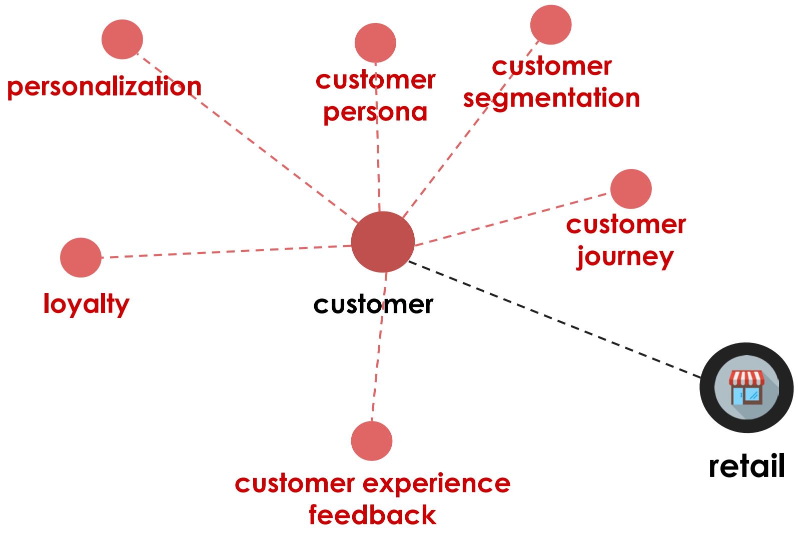
Problem Space - Retail Industry



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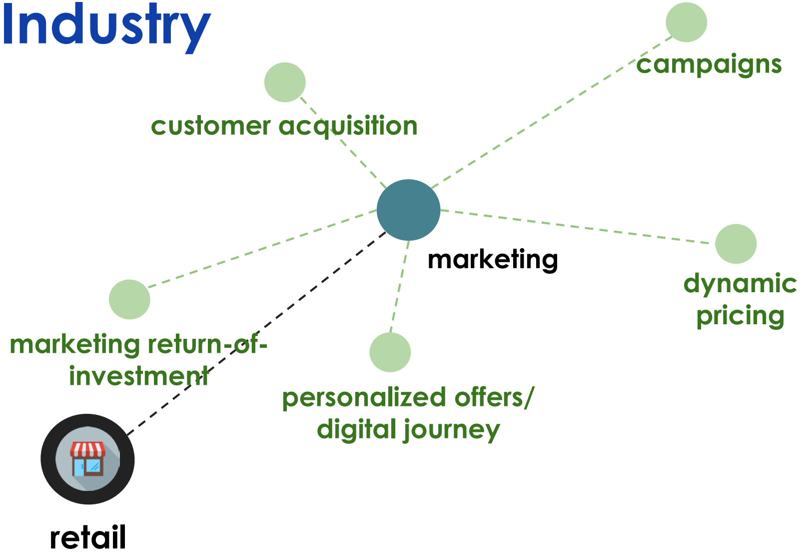
Problem Space - Retail Industry



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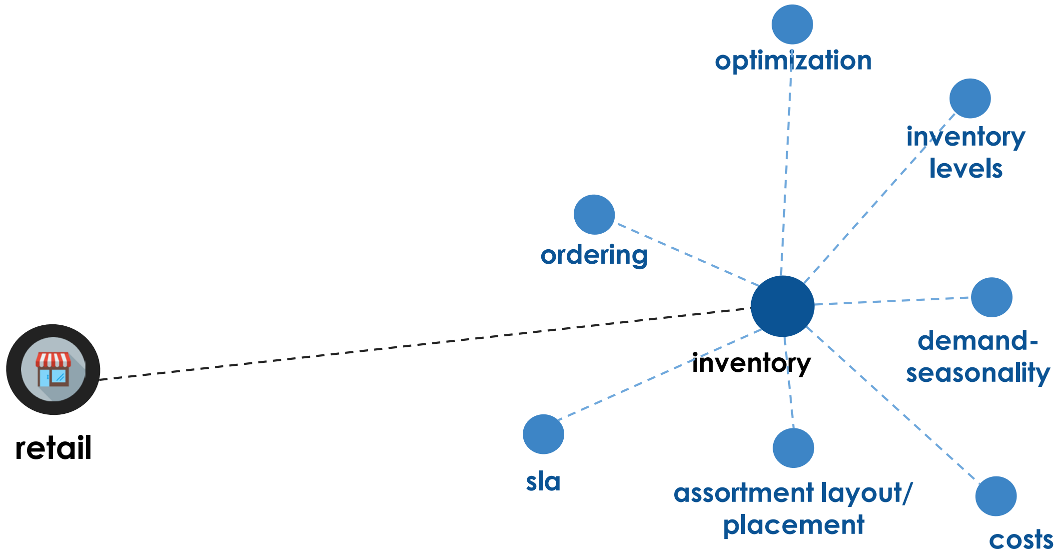
Problem Space - Retail Industry



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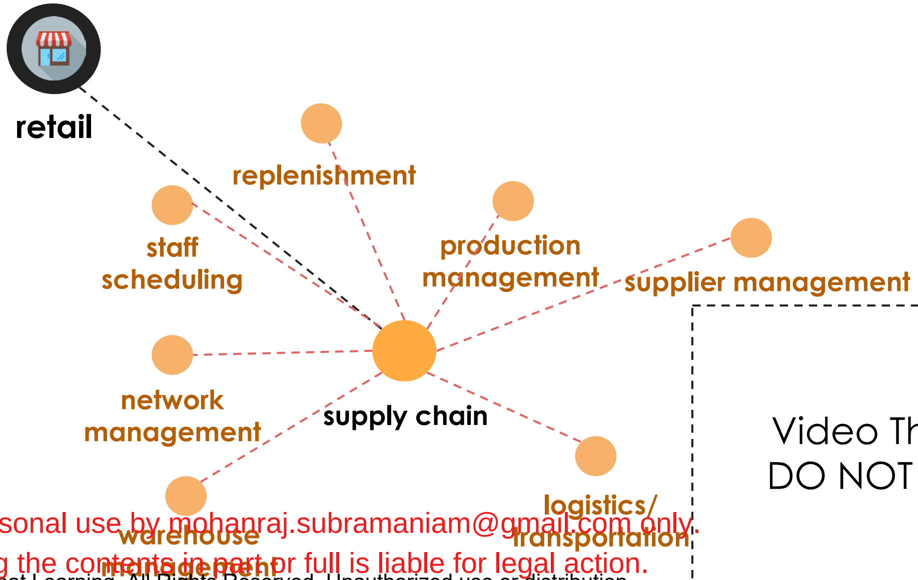

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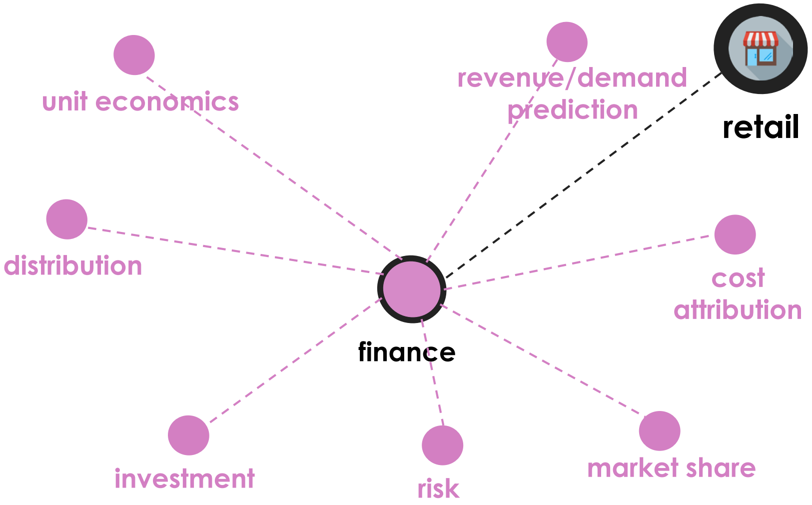

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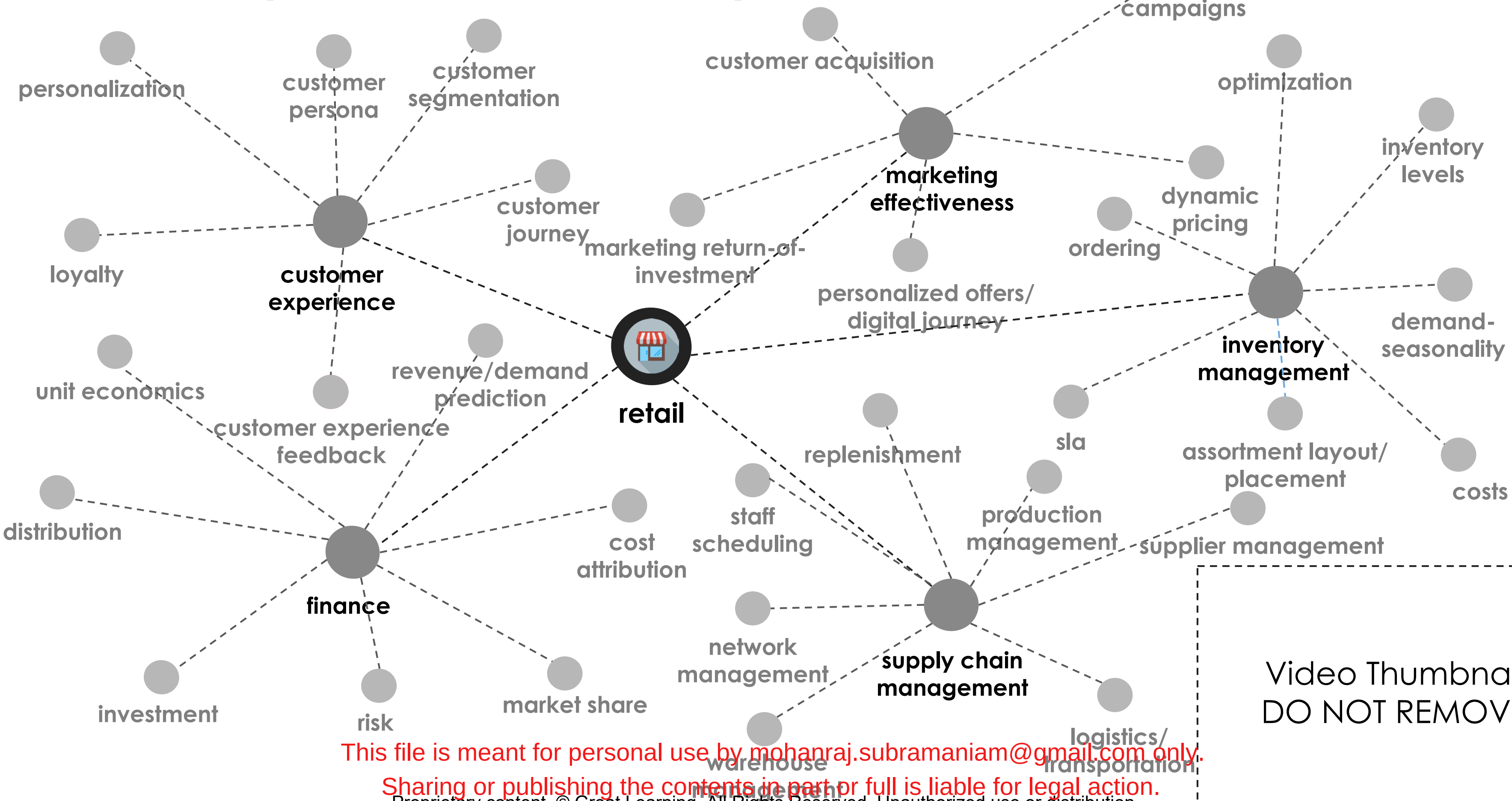

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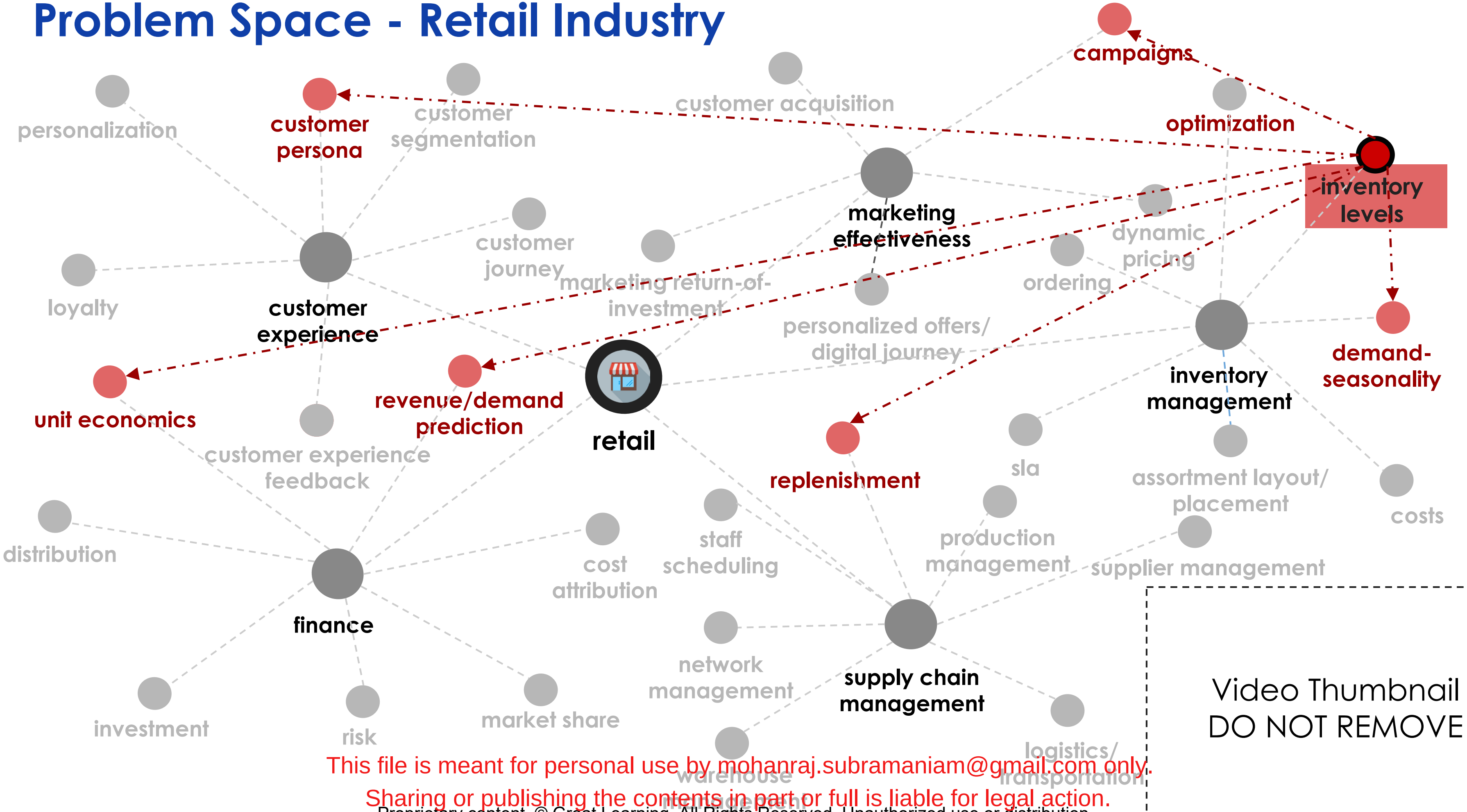

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Optimizing Inventory Levels

Current State → Desired State

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Optimizing Inventory Levels

Current State

- Suboptimal inventory levels
- Loss of revenue due to out of stocks
- Additional cost and wastage due to overstocking
- Suboptimal profits

Desired State

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Optimizing Inventory Levels

Current State

- Suboptimal inventory levels
- Loss of revenue due to out of stocks
- Additional cost and wastage due to overstocking
- Suboptimal profits

Desired State

- Optimal inventory levels
- Minimal revenue losses due to out of stock
- Lower costs related to overstocking
- Better profits and customer satisfaction

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Optimizing Inventory Levels

Current State

- Suboptimal inventory levels
- Loss of revenue due to out of stocks
- Additional cost and wastage due to overstocking
- Suboptimal profits

Gap/Key Questions

- What are the factors that affect the inventory levels?
- What factors are contributing to overstocking and understocking?
- What is the optimal replenishment strategy?

Desired State

- Optimal inventory levels
- Minimal revenue losses due to out of stock
- Lower costs related to overstocking
- Better profits and customer satisfactio

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Optimizing Inventory Levels – data science solution

opt. I ⇒ max. Rev / min. cost ⇒ set smart SLAs + meet them

{ demand forecast + seasonality

cluster

+ replenishment constraint

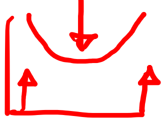
lead time

n/w
warehouse
production.

Safety stock.

$ A.C.

"optimal"



order quantity

"statistics"

EOQ

+ SLA

⇒ $= f(d, c, r.c.)$

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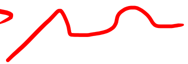
Opt. Inv. ⇒ maximize Rev / minimal costs ⇒ set smart SLAs + meet them

+ variations across time



P/S demand forecast

\+ cluster by product type + cluster by seasonality

Segm/US

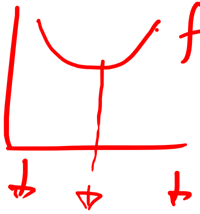
supplier leadtime

US

\+ ~~lead time~~

SLA.

EOQ

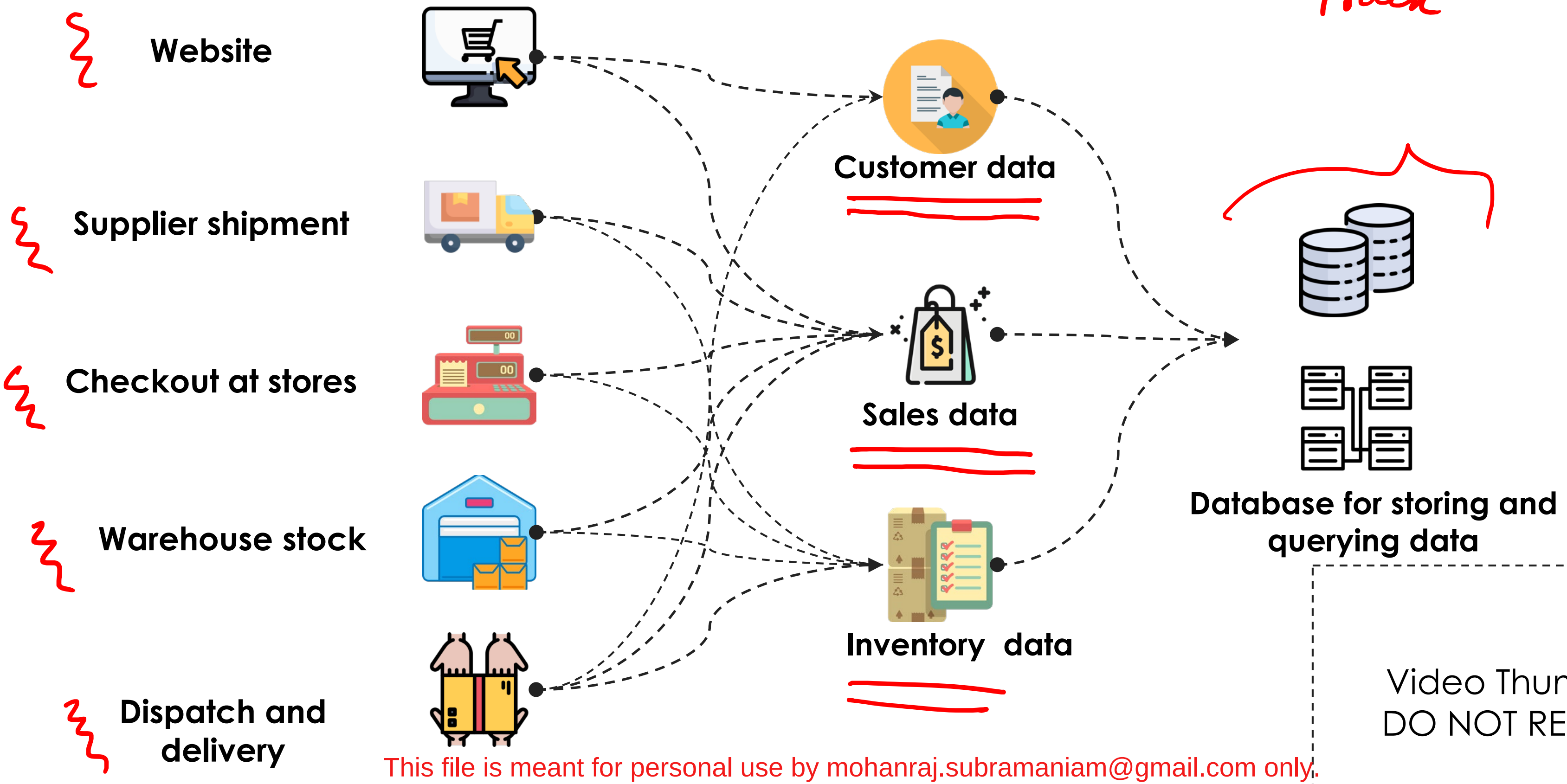


$f(d, l, sla)$

$\overline{LT}$

Optimizing Inventory Levels – Approach - Data

"Track"



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Optimizing Inventory Levels – Approach - Factors

Database for storing and querying data

Product Type	Groceries, Clothing, Electronics, Toys, Grooming
Brand	Apple, Microsoft, Nestle, Heinz, Bosch, Gucci
Supplier	J&J - Neutrogena, Aveeno P&G - Pampers, Tide, Gillette
Seasonality	Winter - Boots, Sweaters Summer - Coolers, Sunscreen
Sales Velocity	Bestsellers vs. slow-moving goods
Sister Products	Clothing + accessories Laptops + chargers

Visualizations using Dashboard

Grouping similar products into categories enables easier tracking of inventory, while maintaining optimal stocks.

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Optimizing Inventory Levels – Approach

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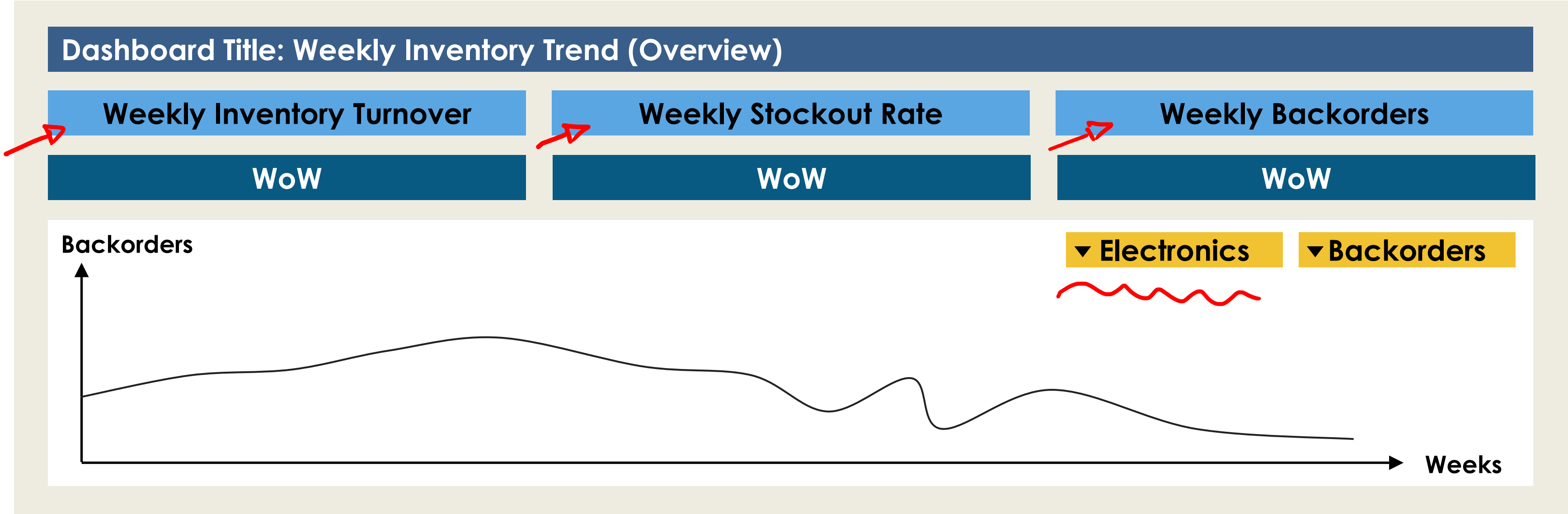
Optimizing Inventory Levels - Tracking



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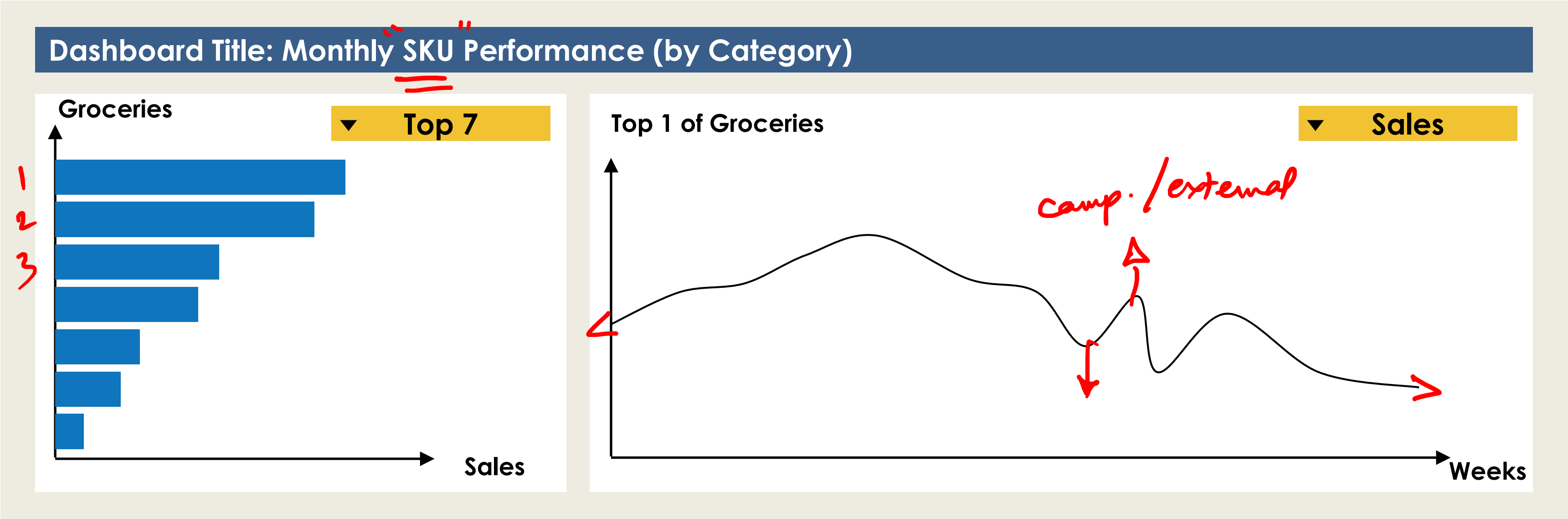
Optimizing Inventory Levels - Tracking



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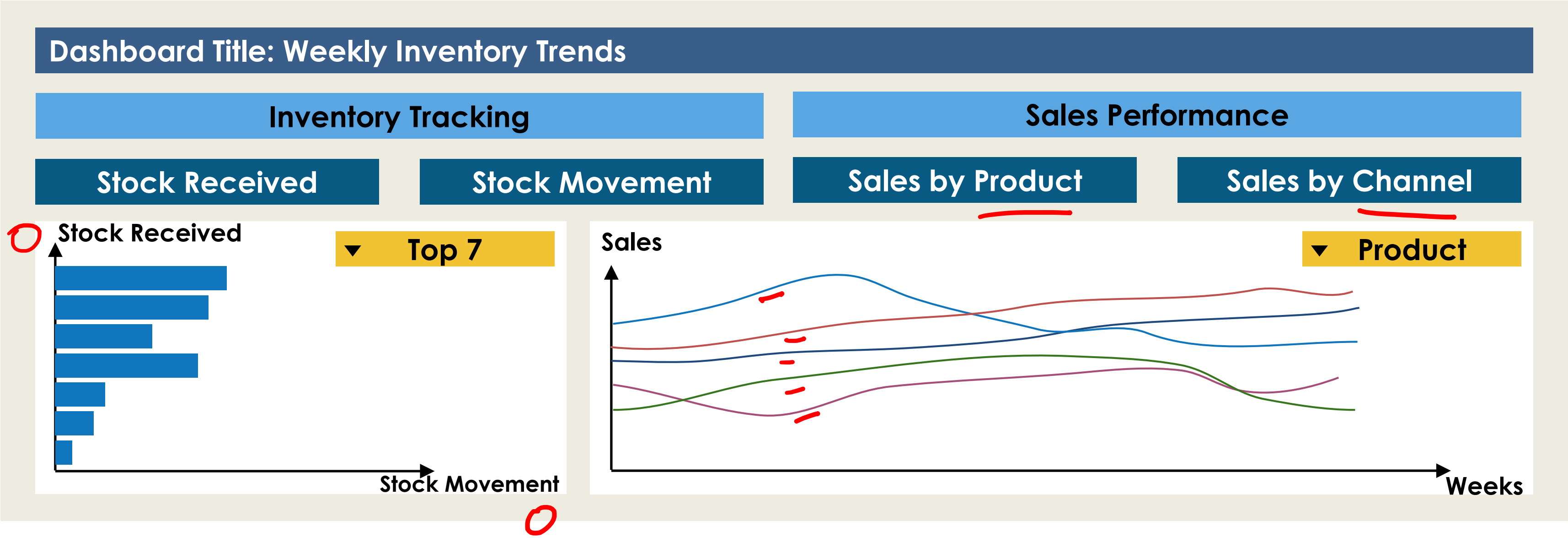
Optimizing Inventory Levels - Tracking



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Problem Space - Healthcare Industry

healthcare

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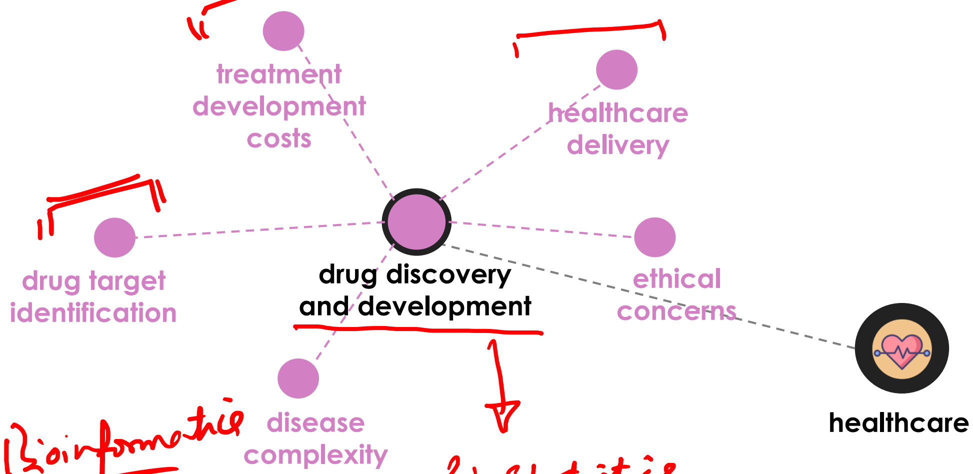
Problem Space - Healthcare Industry



Bioinformatics

Bio Statistics

Clinical Trial

T – Δ – C

Dr

Placebo

Statistics / confidence => Δ

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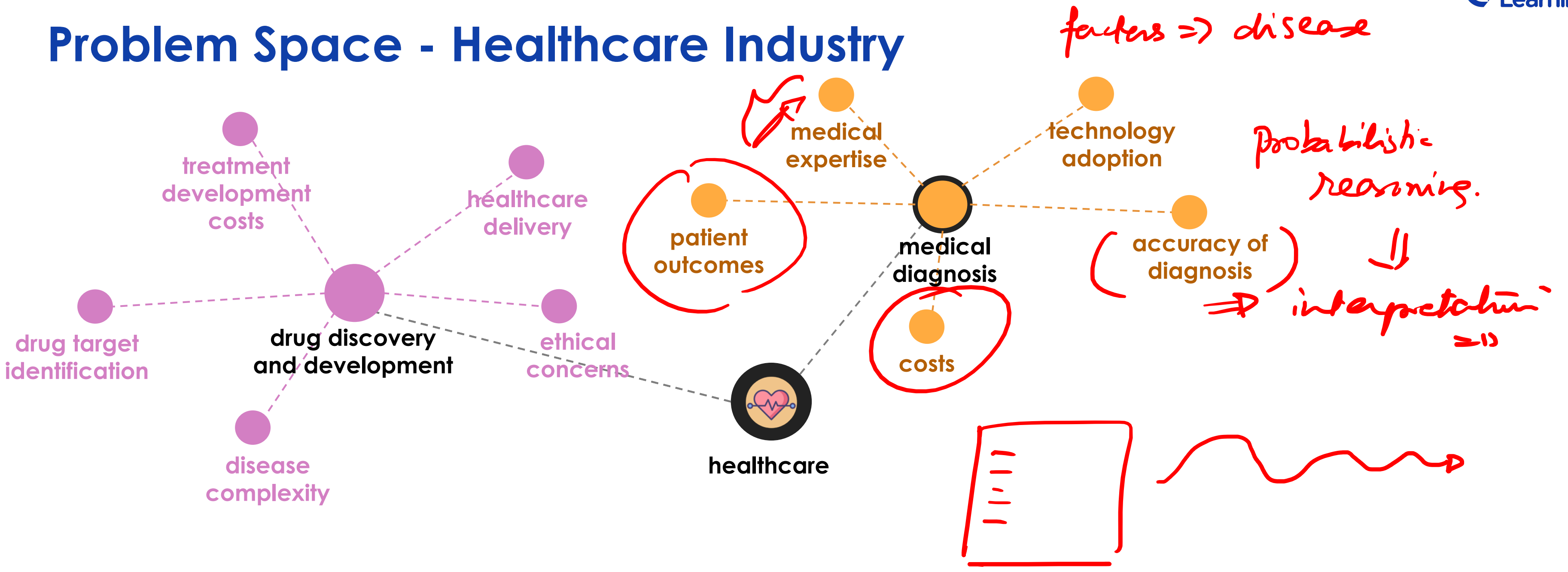
Problem Space - Healthcare Industry



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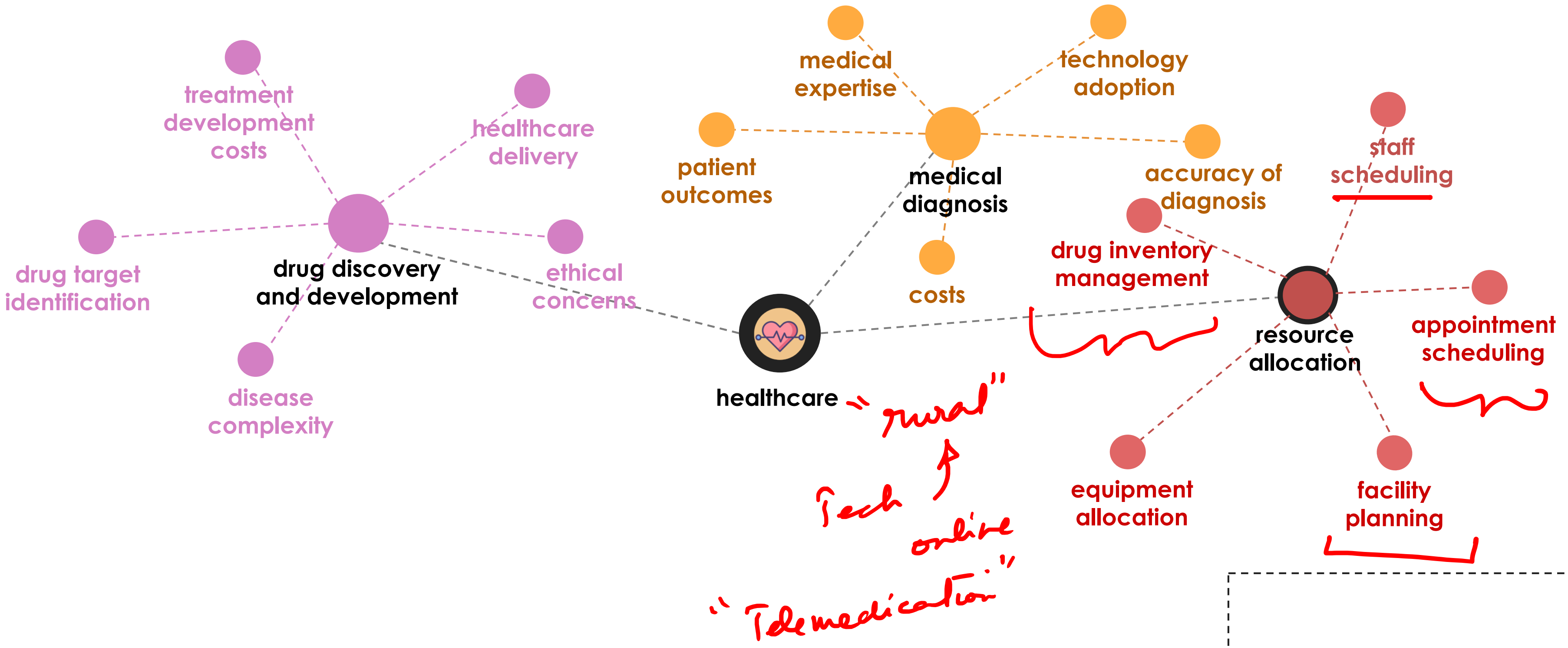
Problem Space - Healthcare Industry



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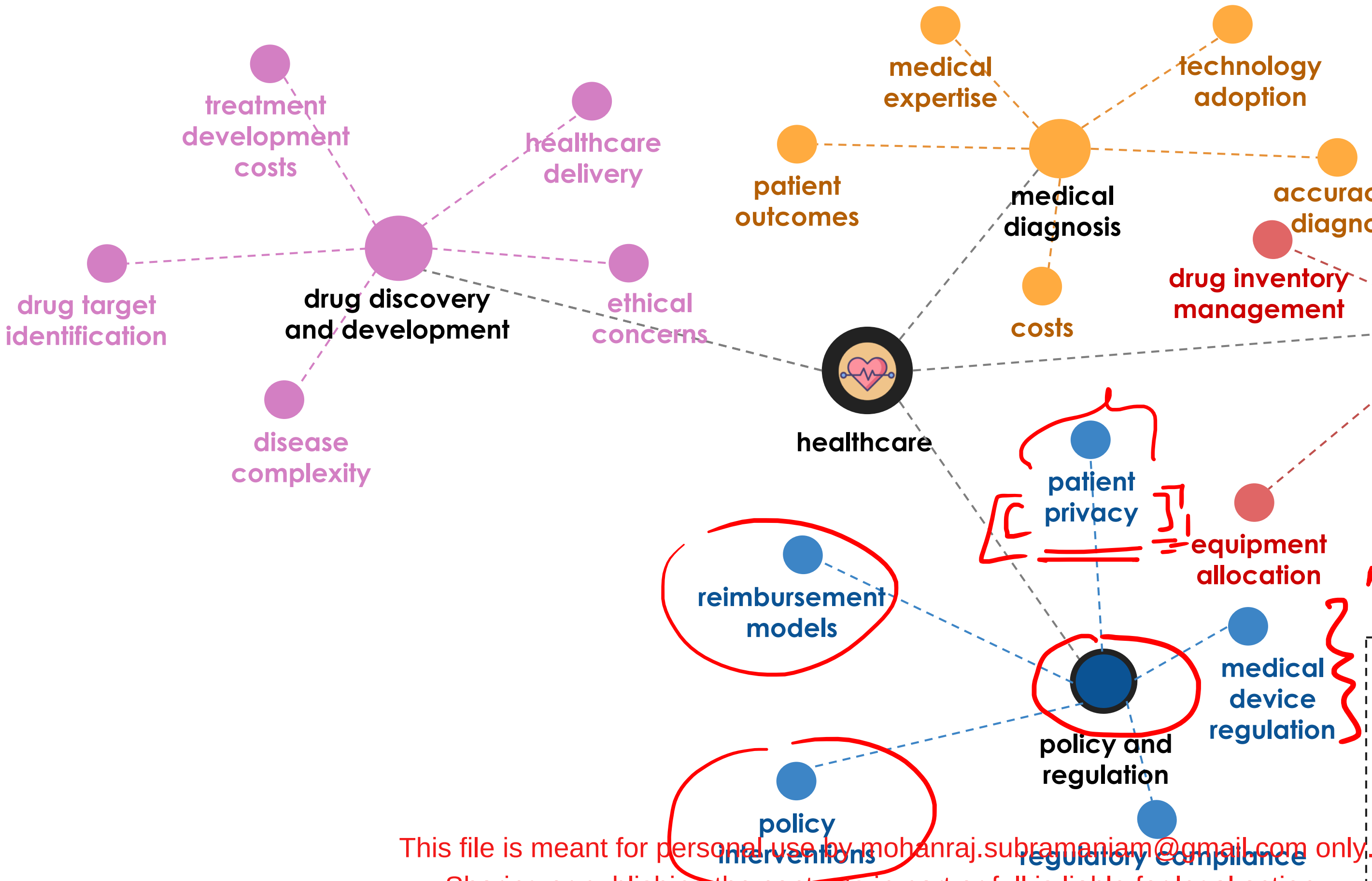

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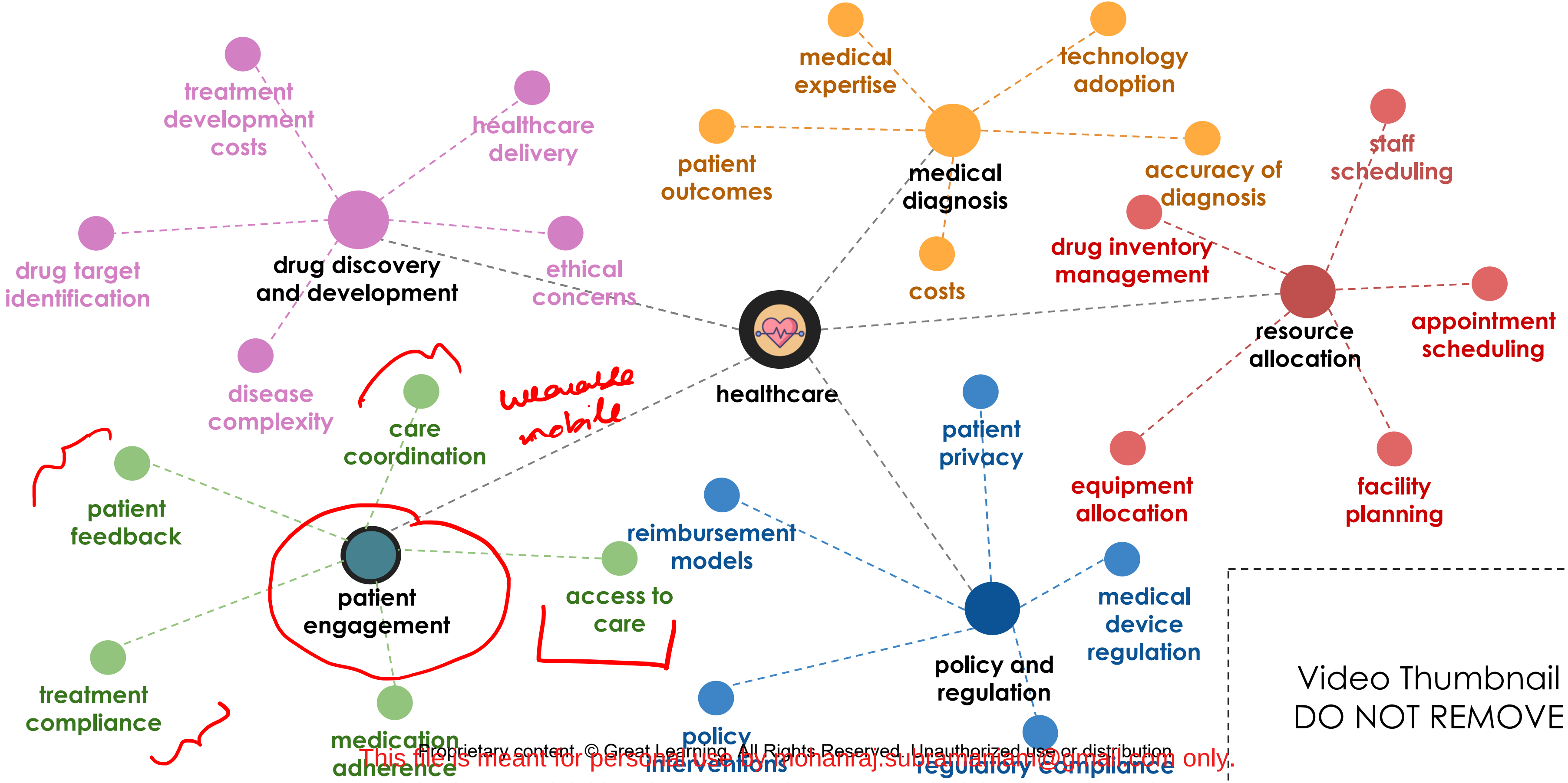

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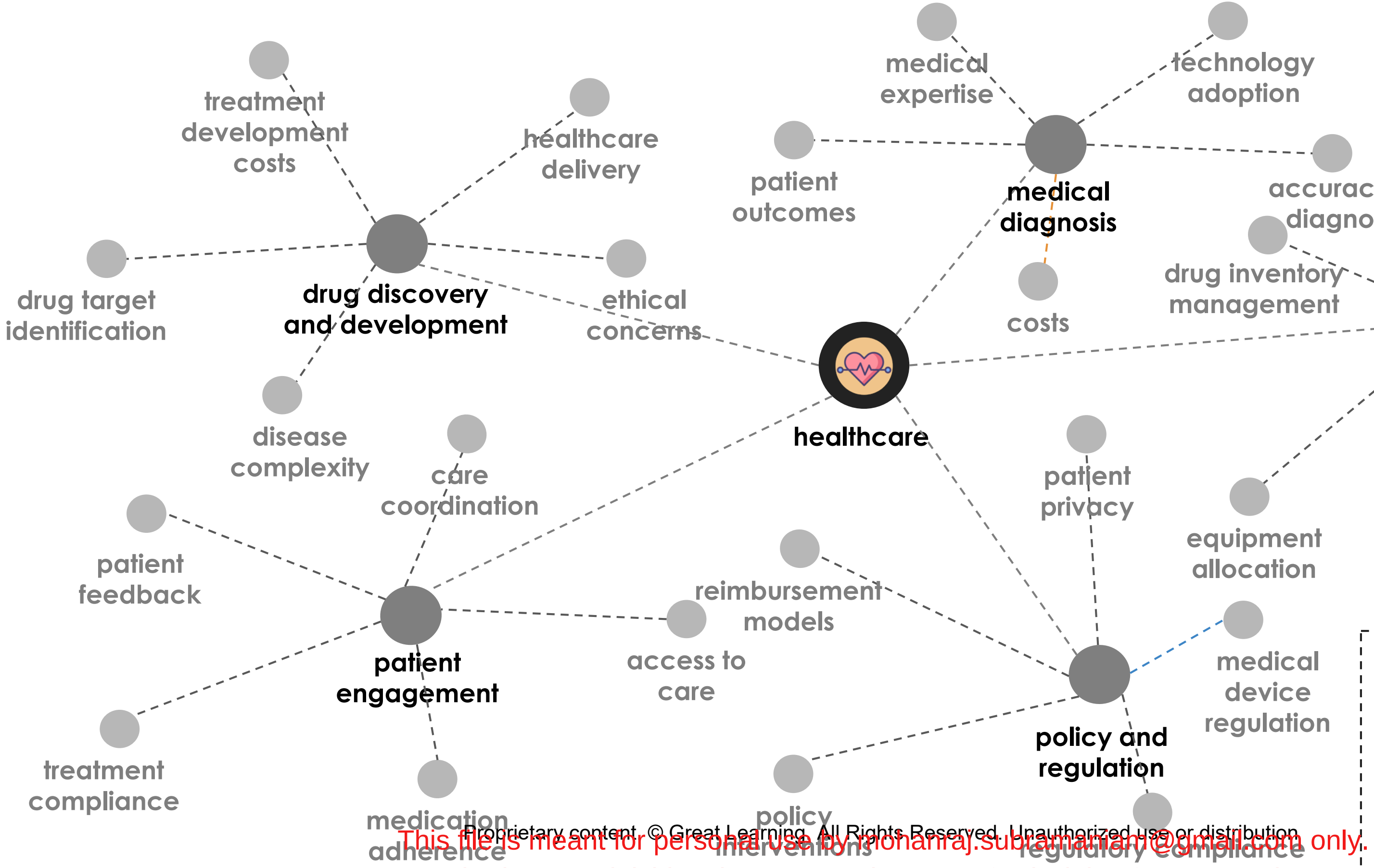

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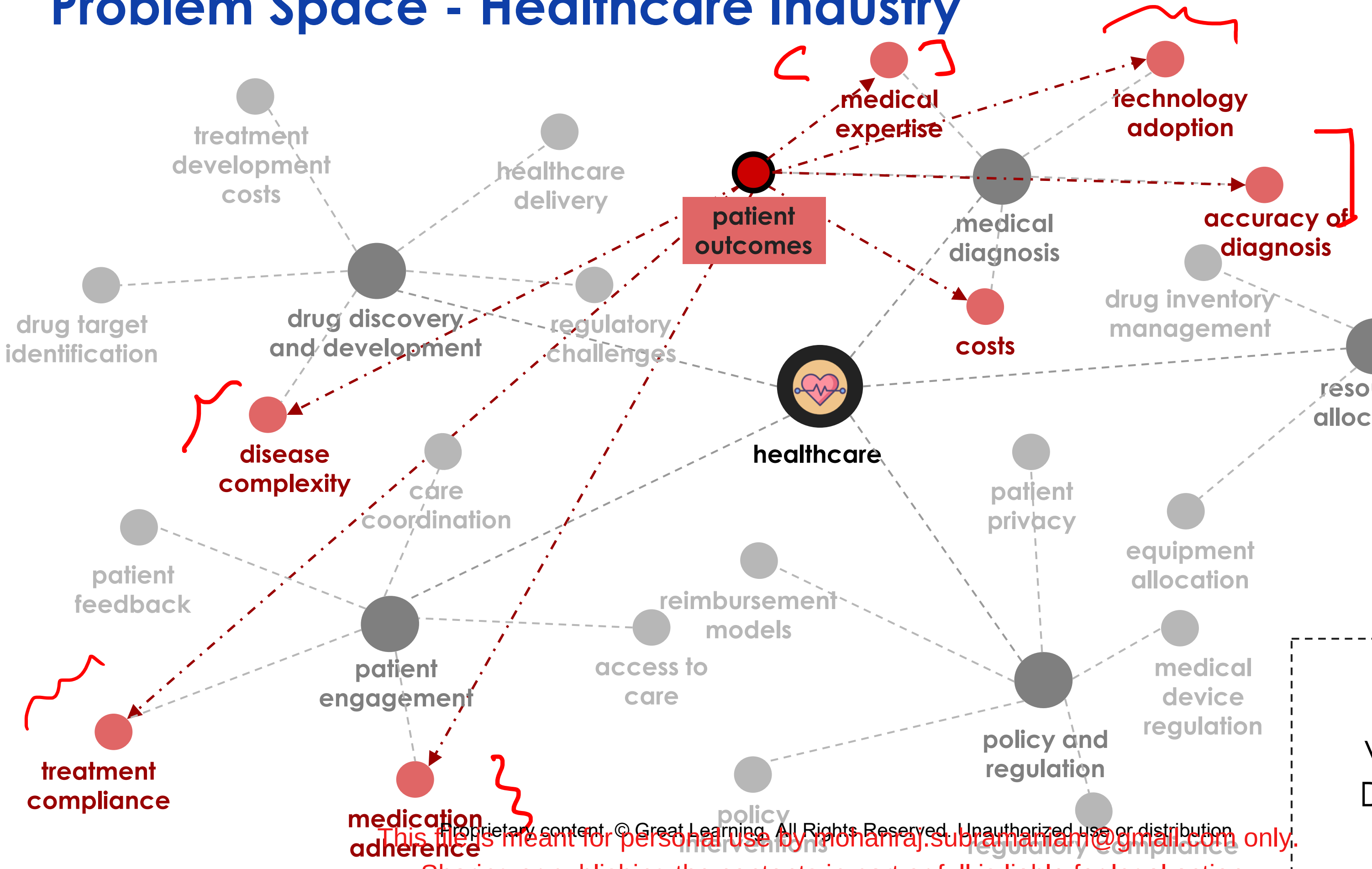

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Assessing the onset of Diabetes

Current State

Gap/Key Questions

Desired State

Absence of screening mechanism to identify patients with a high risk of developing Type 2 Diabetes

Decreased quality of life due to the development of more complications

What are the factors that influence the development of Type 2 Diabetes?

What are the patterns among healthy vs. affected patients?

Identified patterns in patients that can signal the likelihood/ early onset of Type 2 Diabetes

Personalized and immediate intervention to prevent further development of the disease

Improved patient outcomes and reduced healthcare costs

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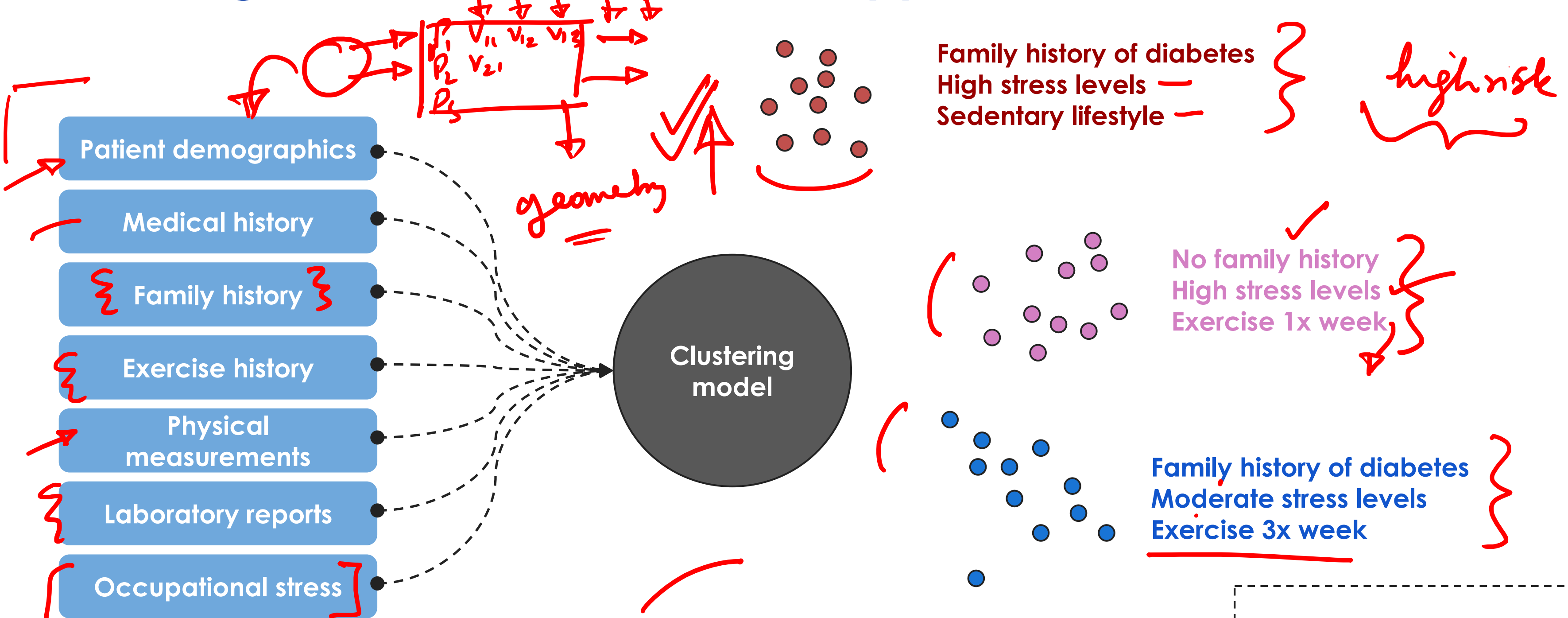
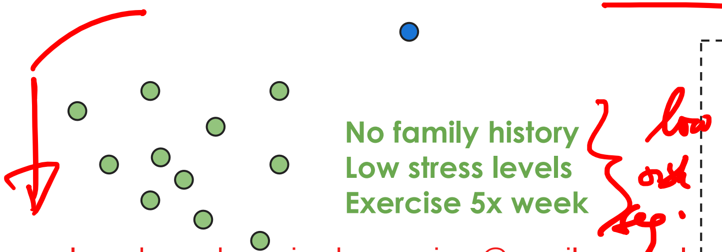
Assessing the onset of Diabetes - Approach





The clustering model groups similar data points together by learning the patterns within the input data.

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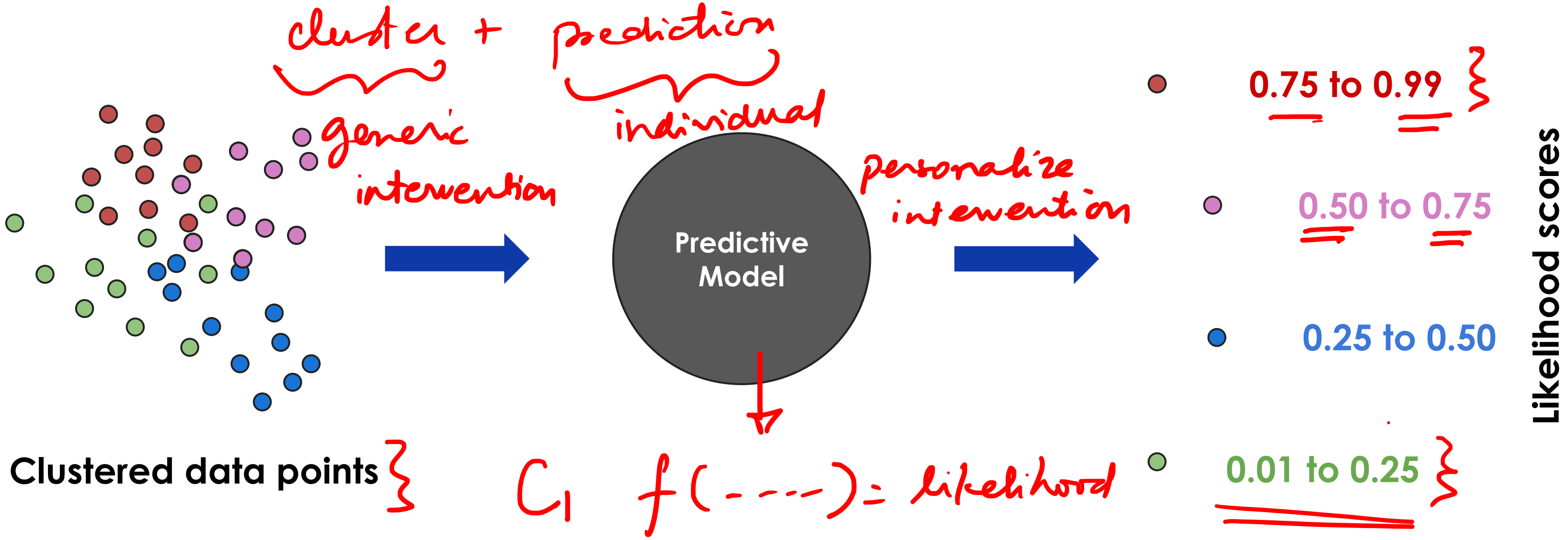
Assessing the onset of Diabetes - Approach



The predictive model assigns a likelihood score to each data point depending on its features and the group characteristics.

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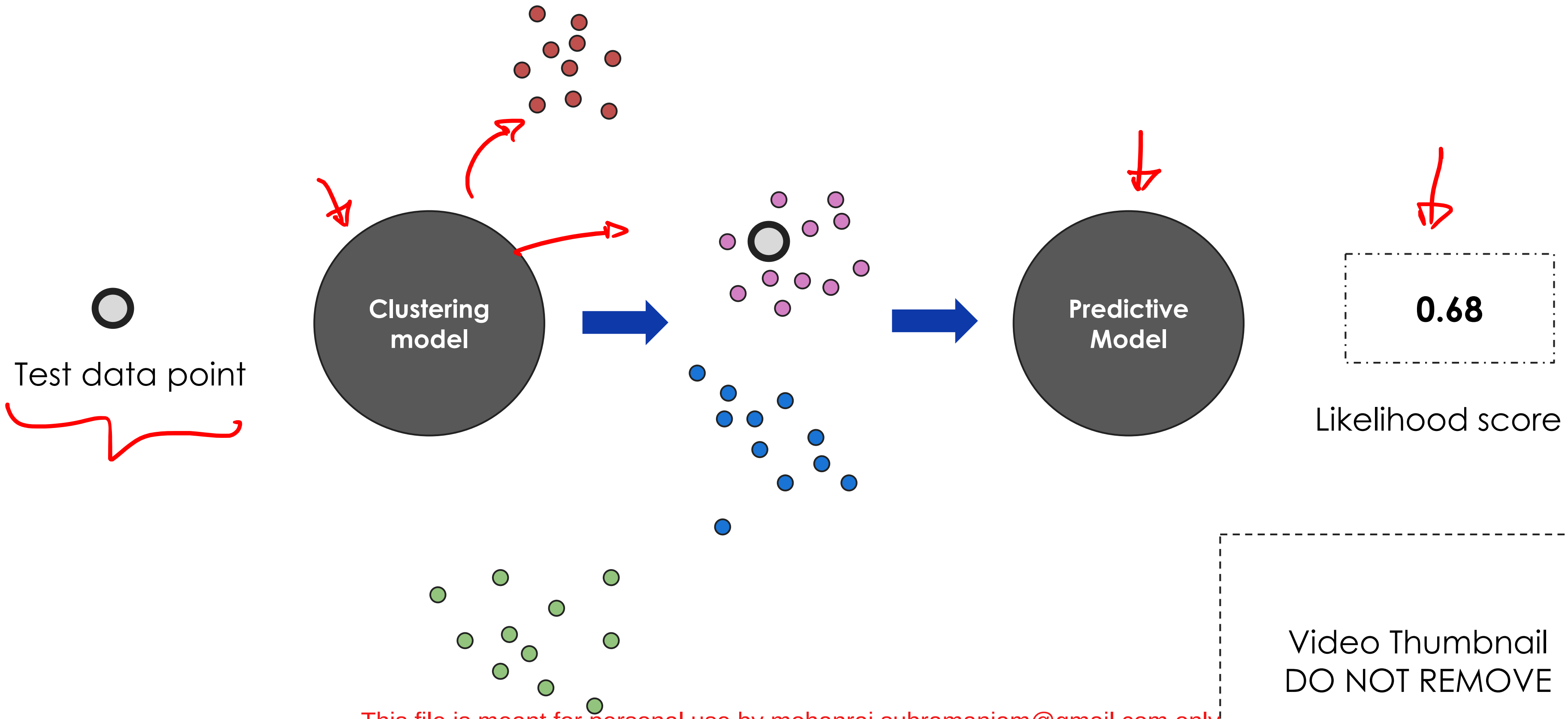
Assessing the onset of Diabetes - Approach



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Problem Space - Banking Industry

banking

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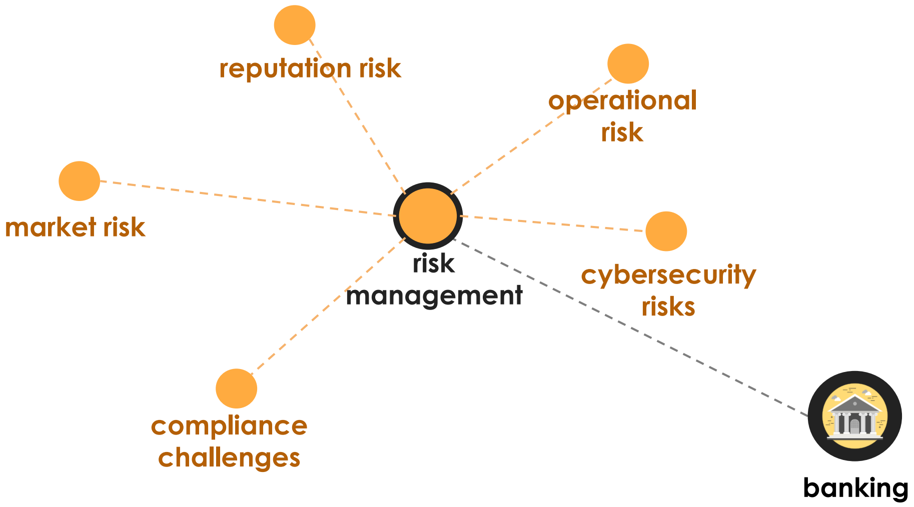
Problem Space - Banking Industry



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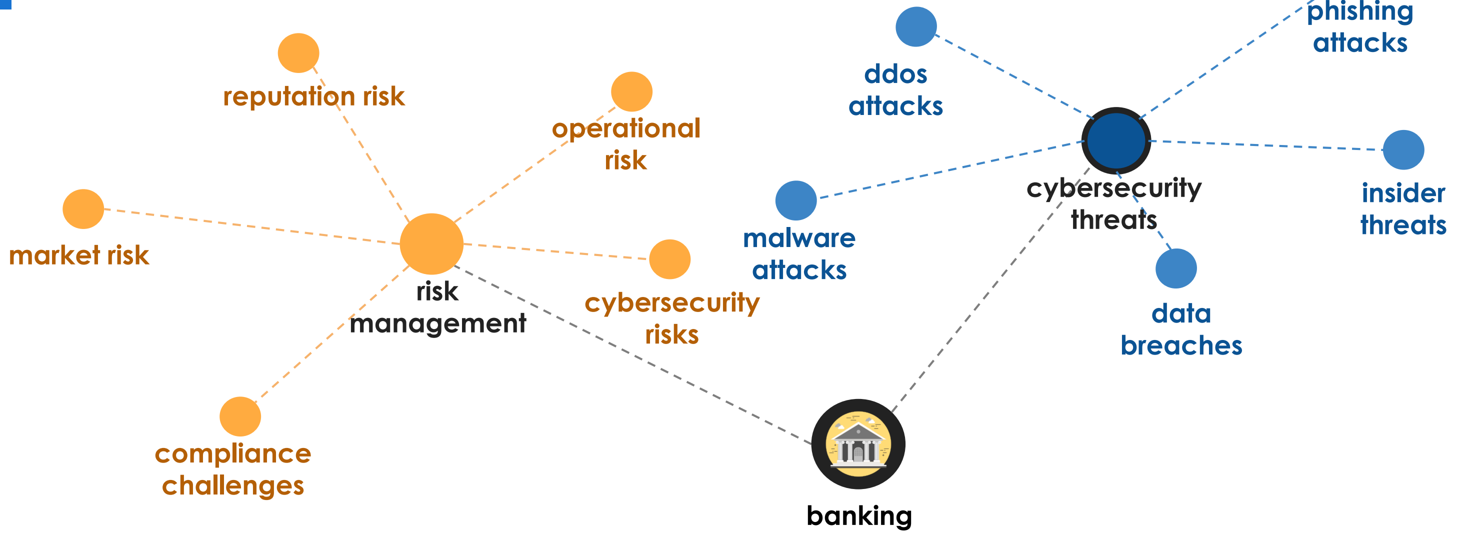
Problem Space - Banking Industry



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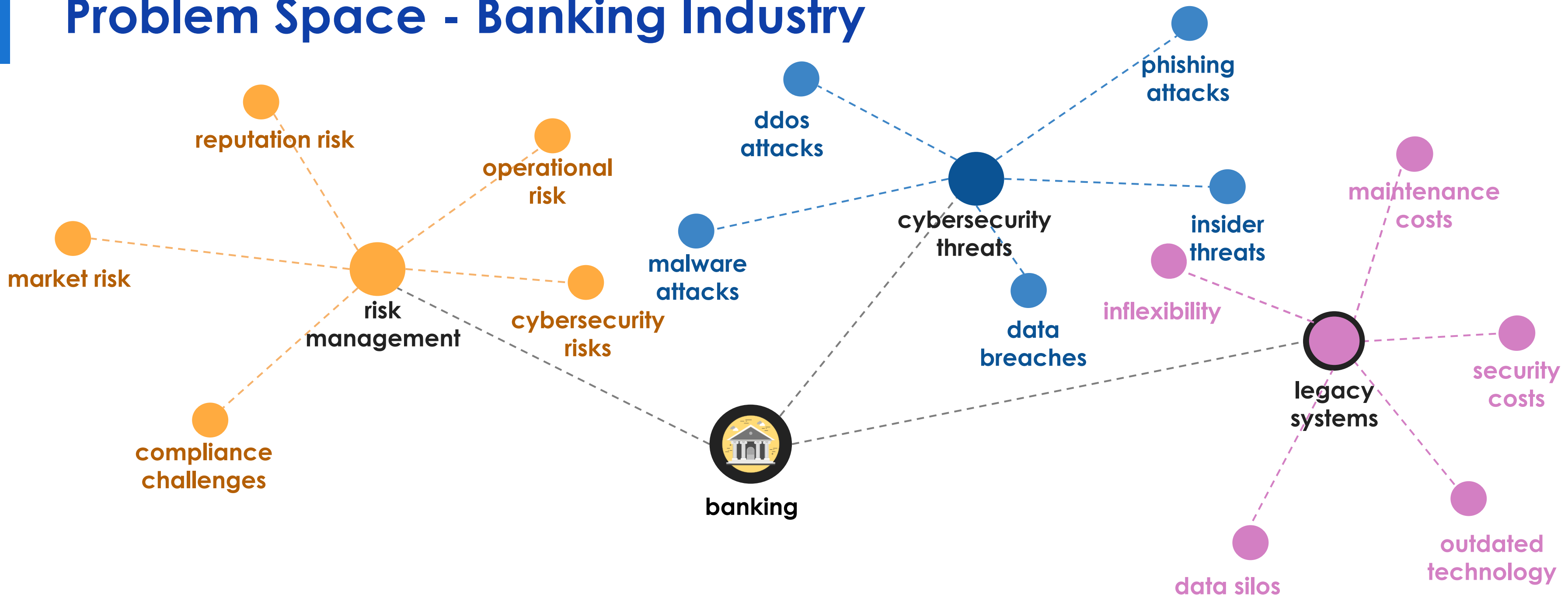
Problem Space - Banking Industry



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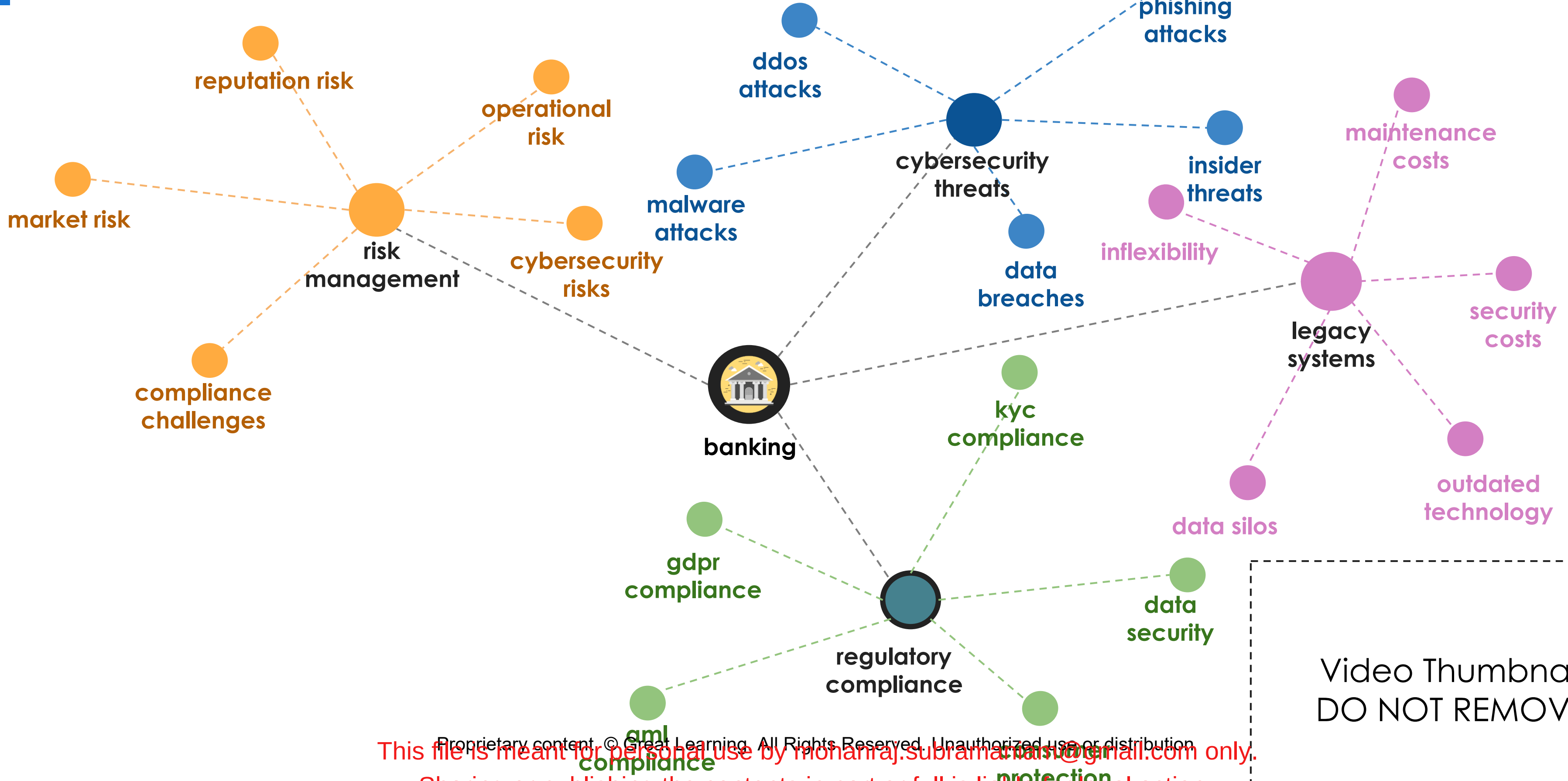

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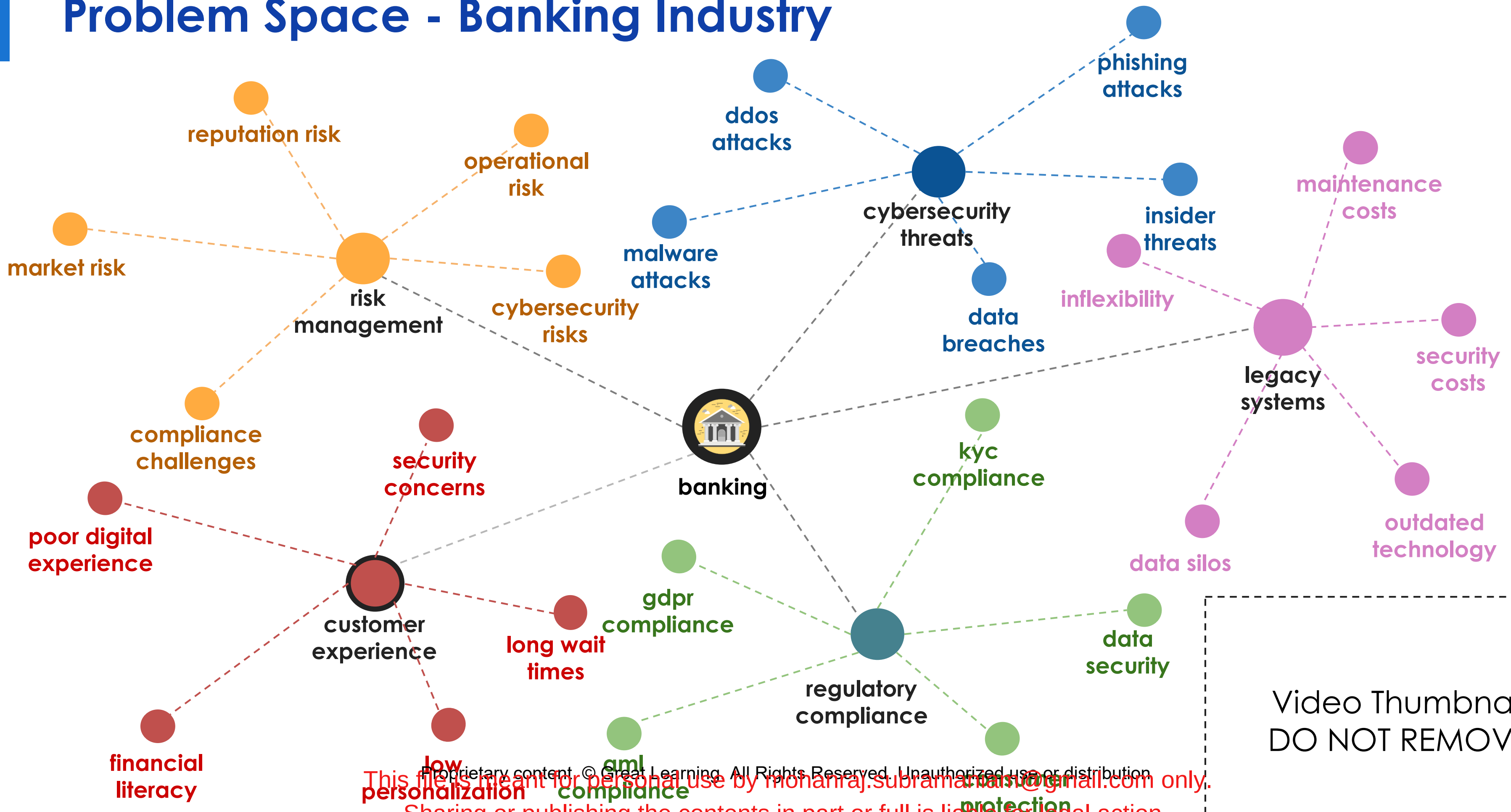

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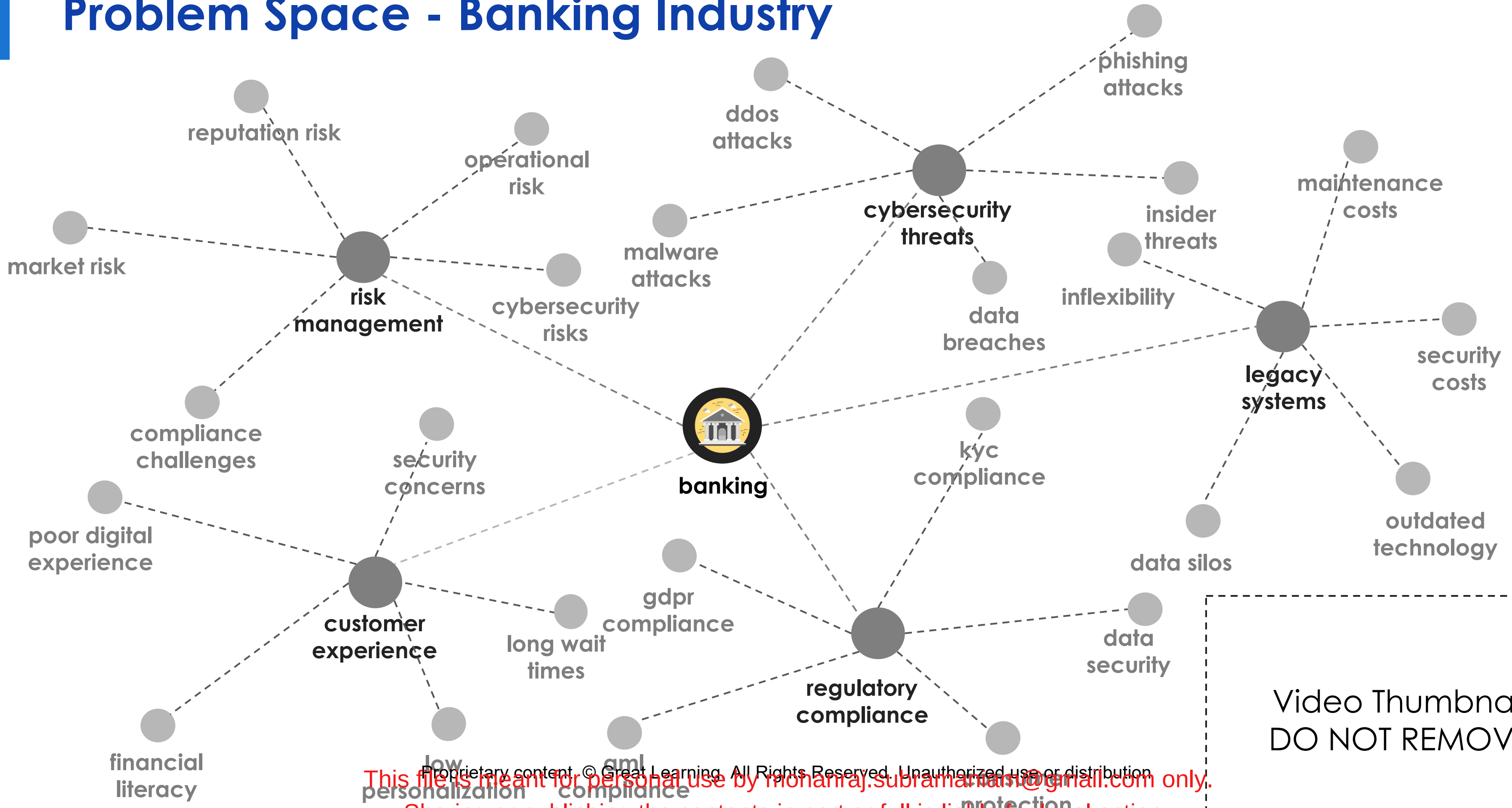

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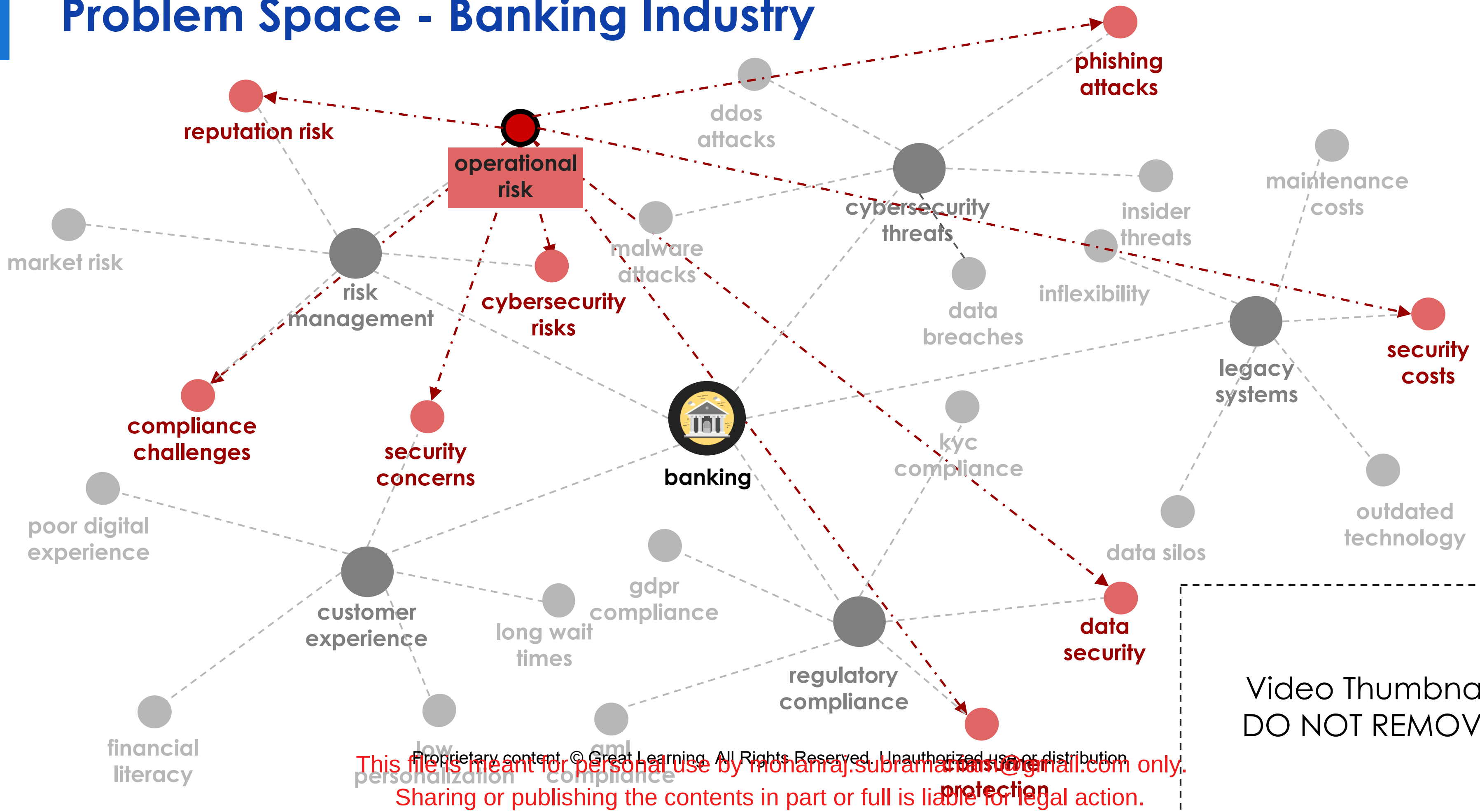

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Credit Card Fraud detection

Current State

Rise in fraudulent transactions by x% YoY

Increase in revenue loss y%

Distressed customer base shrinking YoY

Gap/Key Questions

"data"

What are the key attributes of a fraud transaction?

How do we flag suspicious activity before the transactions are complete?

Desired State

"Minimal fraud transactions << x%

"Minimal loss of revenue due to fraud <<y%

"Growing customer base" due to improved customer satisfaction / trust

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Credit Card Fraud detection - Approach

- Is the transaction amount unusually high?
- Is the location of transaction new/unusual?
- Is the device new/old?
- Is the merchant category new/unusual?
- What kind of transactions have been performed in the past? C M
- When was the card issued?
- Has the card been reported stolen/lost?

→ **Classifier** → **0.83** (Probability score)

Threshold = 0.75

- **Suspicious** (FRAUD)
- **Genuine**

0.8
.07
0.6
.9
.001

cost of acting.

risk score

The classifier learns the features of the input data to classify data into categories based on a certain threshold.

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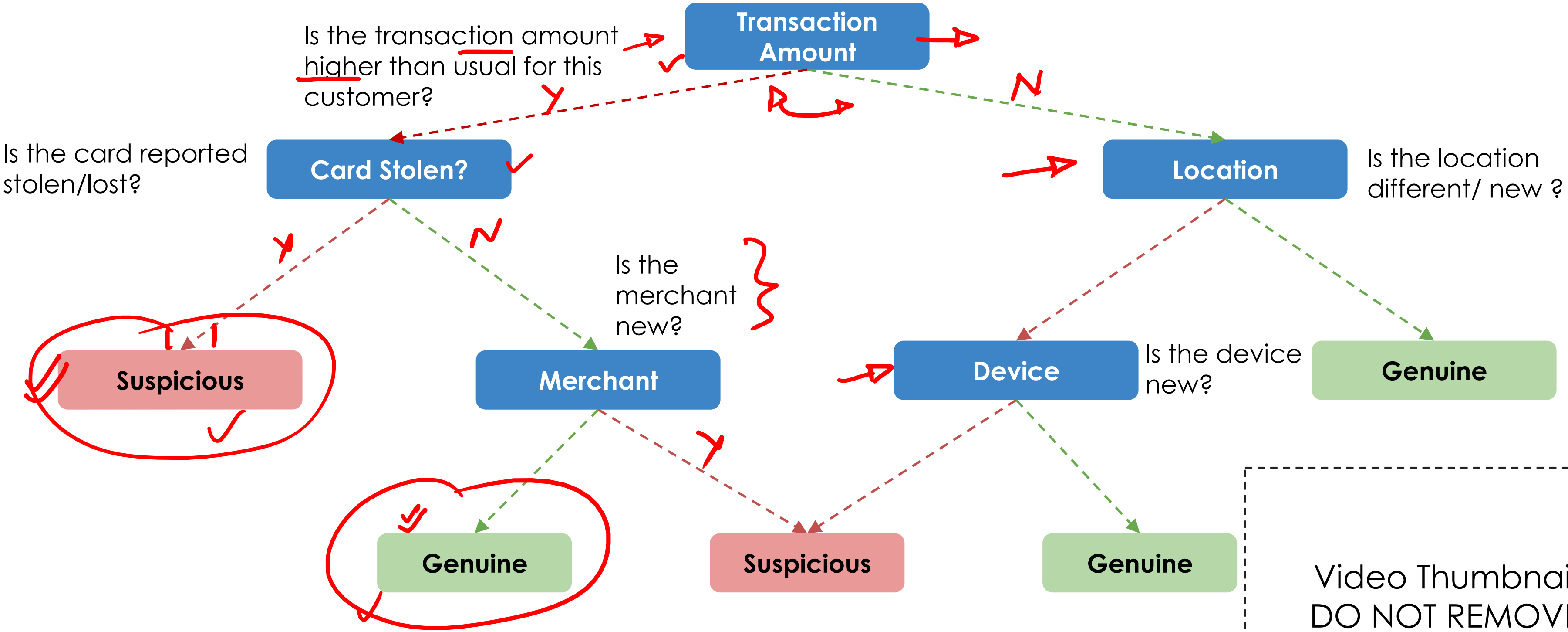
Credit Card Fraud detection - Approach

A simple decision tree

Threshold: More than 2 questions are suspicious



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Summary

- The problems exist in an interconnected manner
- Solving one problem requires one to think thru many factors
- Ability to think through factors is driven by the depth of domain knowledge
- Collecting, preparing and using the right data is directly dependent on domain knowledge
- Using the right data science method is also heavily dependent one's understanding of the domain
- Defining the problem well in light of the domain context is critical to finding right insights and delivering impact for the business

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Happy Learning !